Conjecture 00000003890: infinitely many jumps of one fixed ideal

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1 A fixed ideal and the definition

The conjecture claims that the full set of F-jumping parameters of a fixed ideal is always finite. Let $R = \mathbf{F}_2[X]$ and fix the nonzero proper ideal $I = (X)$. Every positive integer is an F-jumping parameter of I. Thus the first clause is false; no assertion about the second clause, concerning unions inside $(0, 1)$, is needed. We count all positive jumps, as the source states, rather than restricting to a bounded interval.

For $q = 2^s$ with $s \geq 1$, the Frobenius power $J^{[q]}$ is the ideal generated by the qth powers of elements of J. The Frobenius root $A^{[1/q]}$ is the smallest ideal J with $A \subseteq J^{[q]}$. For $t \geq 0$, the generalized test ideal is

$$\tau(I^t) = \bigcup_{s \geq 1} (I^{\lceil tq \rceil})^{[1/q]}, \qquad q = 2^s. \tag{1}$$

This is the standard definition in the regular F-finite polynomial-ring setting; see Definition 4.1 of [1]. The ring $\mathbf{F}_2[X]$ is within that setting. Below we compute actual ideals, and verify that the family in (1) is increasing, so its union agrees with its ideal supremum.

2 The least Frobenius root is computed

For every $N \geq 0$,

$$((X^N))^{[1/q]} = (X^{\lfloor N/q \rfloor}). \tag{2}$$

Indeed, the right-hand ideal has Frobenius power $(X^{q\lfloor N/q \rfloor})$, which contains (X^N). For minimality, let $J = (g)$ be any ideal whose Frobenius power contains X^N. The polynomial ring is a principal ideal domain, and $g^q \mid X^N$. Since X is prime, g is associated to X^j for some $j \geq 0$. The divisibility gives $qj \leq N$, hence $j \leq \lfloor N/q \rfloor$. Therefore $(X^{\lfloor N/q \rfloor}) \subseteq J$, proving minimality against every candidate ideal, not only against other monomial ideals.

Formula (2) identifies the sth ideal in (1) as $(X^{\lfloor \lceil tq \rceil / q \rfloor})$. The inequality $\lceil 2tq \rceil \leq 2\lceil tq \rceil$ shows that these ideals increase with s. They decrease when t increases. At a nonnegative integer k, every ideal in (1) is (X^k), giving

$$\tau(I^k) = (X^k). \tag{3}$$

3 Every positive integer is a jump

For $k \geq 0$ and $q = 2^s$, define

$$t_{k,s} = \frac{kq + q - 1}{q} = k + 1 - \frac{1}{q}.$$

Then $0 \leq t_{k,s} < k + 1$ and $t_{k,s} \to k + 1$ from the left. The ideal in (1) at index s is (X^k), since $\lceil t_{k,s} q \rceil = kq + q - 1$ and $\lfloor (kq + q - 1)/q \rfloor = k$. Consequently

$$X^k \in \tau(I^{t_{k,s}}), \qquad X^k \notin \tau(I^{k+1}) = (X^{k+1}).$$

For any $0 \leq t < k+1$, choose s large enough that $t < t_{k,s}$. The decreasing dependence of $\tau(I^t)$ on t then implies $\tau(I^t) \neq \tau(I^{k+1})$. This proves the usual left-jump criterion at every $k+1$, and in particular rules out local constancy from the left. The injective map $k \mapsto k+1$ from $\mathbf{N}$ into the jump set proves that this fixed ideal has infinitely many jumps.

4 Formalization and verification

The Lean ring is the actual `Polynomial (ZMod 2)`. Frobenius powers use `Ideal.map` of `iterateFrobenius`. The root is an `sInf` over all candidate ideals, and `frobeniusRoot_X_pow_least` proves that the infimum is attained and is the least candidate. Thus existence or minimality of roots is not assumed.

`testStages_monotone` proves the increasing chain, and `mem_testIdeal_iff` identifies the ideal supremum with the literal union. `testIdeal_integer` and `testIdeal_antitone` prove (3) and parameter monotonicity. The left-approaching points and their monomial witnesses are explicit. `positive_integer_jump` uses the criterion $\tau(I^t) \neq \tau(I^c)$ for every $0 \leq t < c$. Finally `infinitely_many_jumps` is a `Set.Infinite` theorem about the real jump parameters.

The pinned project uses Lean 4.33.0 and exact Mathlib/dependency commits. Independent project build, fifteen-declaration axiom audit and official fresh-kernel replay are recorded. No admitted proof, native decision shortcut or extra axiom is used. The original bilingual source is fixed at upstream commit
`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

OpenAI Codex assisted the proof and report; a separate supervising Codex agent reviewed the original, definitions and full formal chain. This is AI review, not human certification. The example is classical.

References

[1] W. Badilla-Céspedes and E. León-Cardenal, *F-Pure Thresholds and F-Volumes of Some Non-Principal Ideals*, Mediterranean Journal of Mathematics 22, 98 (2025), Definition 4.1. `https://arxiv.org/html/2305.00571v3`.